

Project Interim Report
On
AUTOMATIC CORE SAMPLE LOGGING
(Using Python & ML)
RU-500-02-00002
Foundations of Machine Learning



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TECHNICAL REPORT

Automatic Core Sample Logging & Mineral Identification System

Machine Learning-Based Drill Core Analysis Using Computer Vision and Deep Learning

1. Executive Summary

This report documents an automatic core sample logging system that leverages machine learning and computer vision to identify and classify minerals from drill core and rock sample images. The system is designed to automate what has traditionally been a manual, time-intensive geological process, offering geologists a rapid, scalable, and objective first-pass analysis tool.

The pipeline integrates classical image processing, unsupervised clustering, a deep learning classifier (ResNet50), and an interactive Gradio web interface. It supports seven common minerals and provides per-sample logging with color, hardness, texture, and geological context.

2. Objectives

- Automate mineral identification from core sample or rock photographs.
- Extract and analyze visual features (color, texture, structure) for classification.
- Provide a trained deep learning model (ResNet50) for high-accuracy mineral recognition.
- Deliver an interactive, browser-based interface for field or lab use.
- Generate structured logging summaries suitable for geotechnical records.

3. System Architecture

The system is structured as a sequential pipeline implemented in Python within a Jupyter Notebook (Google Colab environment). The architecture consists of five major layers:

3.1 Environment & Dependency Layer

Core libraries include TensorFlow/Keras for deep learning, OpenCV and scikit-image for image processing, scikit-learn for classical ML, and Gradio for the user interface. All dependencies are installed via pip at runtime.

3.2 Data Ingestion Layer

Users upload labelled image datasets (individual images or ZIP archives) directly into the Colab environment. The system expects a directory structure organized by mineral class (e.g., dataset/train/quartz/, dataset/train/pyrite/), compatible with Keras ImageDataGenerator.

3.3 Feature Extraction & Preprocessing

Each image is resized to 224x224 pixels, converted to RGB, and normalized to [0, 1]. Two feature extraction methods are employed:

- Color Features: HSV histogram (8 bins on the Hue channel) captures dominant color signature.
- Texture Features: Gray-Level Co-occurrence Matrix (GLCM) contrast metric via scikit-image characterizes surface texture.

3.4 Classification Layer

Two classification approaches are implemented in parallel:

- Unsupervised Clustering (K-Means): Applied to up to 100 sample images to discover visual groupings without labels. The cluster count k is set dynamically (max 5), useful for exploratory analysis.
- Supervised Deep Learning (ResNet50): A pre-trained ResNet50 base (ImageNet weights) is fine-tuned with a GlobalAveragePooling layer, a 128-unit dense layer (ReLU), and a 7-class softmax output. The model is compiled with Adam optimizer and categorical cross-entropy loss.
- Heuristic Fallback: When the CNN has not yet been trained on labelled data, HSV mean values drive rule-based mineral assignment, providing a baseline prediction without model weights.

3.5 Explainability Layer (Grad-CAM)

A Gradient-weighted Class Activation Mapping (Grad-CAM) module generates heatmaps that highlight the image regions most influential for the model's prediction. This is visualized as a side-by-side comparison of the original image and the overlaid heatmap, aiding geologist interpretation and model trust.

3.6 User Interface Layer

A Gradio web interface provides single-image upload, instant mineral prediction, property display, and a structured core logging summary output. The interface is accessible via a shareable public URL (share=True) for remote or field use.

4. Supported Minerals & Reference Database

The system identifies seven minerals relevant to common ore deposit settings. The internal database stores key properties used for logging output:

Mineral	Color	Hardness (Mohs)	Texture	Common In
Biotite	Black/dark brown	2.5–3	Platy, micaceous cleavage	Granites, schists, gneisses
Bornite	Copper-red to purple iridescent	3	Massive, tarnished iridescent	Copper ore deposits, porphyry Cu
Chrysocolla	Blue-green	2–4	Earthy, amorphous	Oxidized copper zones
Malachite	Bright green	3.5–4	Banded, botryoidal	Oxidized copper ore zones
Muscovite	Silver/pale yellow	2–2.5	Pearly sheen, thin flexible sheets	Granites, schists, pegmatites
Pyrite	Brassy yellow	6–6.5	Cubic crystals, striated faces	Sulfide ore deposits
Quartz	White/gray/colorless	7	Glassy, conchoidal fracture	Veins, host rock, sandstone

5. Classification Logic

5.1 Color-Heuristic Baseline

The heuristic classifier analyzes mean Hue (H), Saturation (S), and Value (V) from the HSV color space. Decision rules are:

- Low saturation + high brightness → Quartz (white/gray)
- Low saturation + low brightness → Biotite (dark black/brown)
- Hue 10–25, high saturation → Bornite (copper-red)

- Hue 35–90, high saturation → Malachite (bright green) or Chrysocolla (blue-green)
- Hue 85–130, moderate saturation → Chrysocolla
- Hue 20–35, high saturation, bright → Pyrite (brassy yellow)
- Low-moderate saturation, mid-brightness → Muscovite (silvery)

5.2 CNN-Based Prediction (Post-Training)

Once the ResNet50 model is trained on labelled mineral images, the heuristic fallback is replaced by the neural network output. The softmax probabilities across all 7 classes are used for the final prediction, with confidence values available for uncertainty quantification.

6. Output Format & Logging

For each uploaded image, the system produces three output fields in the Gradio interface:

- Predicted Mineral: The identified mineral name in uppercase.
- Mineral Properties: A formatted block listing color, Mohs hardness, texture, and geological occurrence.
- Core Logging Summary: A structured log entry including HSV analysis values, color contrast metric, and the predicted mineral's full properties — suitable for inclusion in geotechnical reports.

Example log output format:

Identified Mineral : PYRITE		Color: Brassy yellow		Hardness: 6-6.5
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7. Limitations & Recommendations

7.1 Current Limitations

- The heuristic baseline is sensitive to lighting conditions and camera white balance; images with unusual illumination may yield incorrect mineral assignments.
- K-Means clustering does not use class labels, so clusters do not map 1:1 to mineral types without manual inspection.
- The CNN model requires a labelled training dataset of core images; the notebook does not include pre-trained mineral-specific weights.

- Only seven minerals are currently supported; expansion requires additional labelled data and retraining.

7.2 Recommended Improvements

- Acquire and annotate a domain-specific drill core image dataset (e.g., from Kaggle mineral datasets or proprietary core libraries) and train the ResNet50 fine-tuned model to replace the heuristic baseline.
- Add image augmentation (rotation, flipping, brightness jitter) during training to improve model robustness across lighting and orientation variations.
- Extend the mineral database to include gangue minerals and ore minerals common to regional geology.
- Integrate depth tracking metadata (core box ID, depth interval) into the logging output to produce full drill hole composite logs.
- Deploy as a containerized web service (Docker + FastAPI) for production field use without Colab dependency.

8. Technology Stack

Component	Technology	Purpose
Deep Learning	TensorFlow / Keras + ResNet50	CNN mineral classifier
Classical ML	scikit-learn (SVM, RF, K-Means)	Clustering & alternative classifiers
Image Processing	OpenCV, scikit-image (GLCM)	Preprocessing & feature extraction
UI / Deployment	Gradio	Browser-based inference interface
Data Handling	NumPy, Pandas, Matplotlib	Array ops, logging, visualization
Environment	Google Colab / Jupyter Notebook	Interactive development platform
Explainability	Grad-CAM (custom implementation)	Model prediction heatmaps

9. Conclusion

The Automatic Core Sample Logging system represents a practical application of computer vision and deep learning to geoscience workflows. By combining classical HSV-based heuristics with a fine-tunable ResNet50 classifier, the system provides both an immediate out-of-box baseline and a high-accuracy path once trained on domain data.

The Gradio interface lowers the barrier to adoption, enabling geologists without ML expertise to obtain structured mineral logs from core photographs in real time. With

recommended enhancements — particularly labelled training data and depth metadata integration — this system can serve as a cost-effective first-pass logging tool in mineral exploration programs.